Scale Matters: Adaptive Granularity Selection for Cross-Species 3D Plant Organ Segmentation

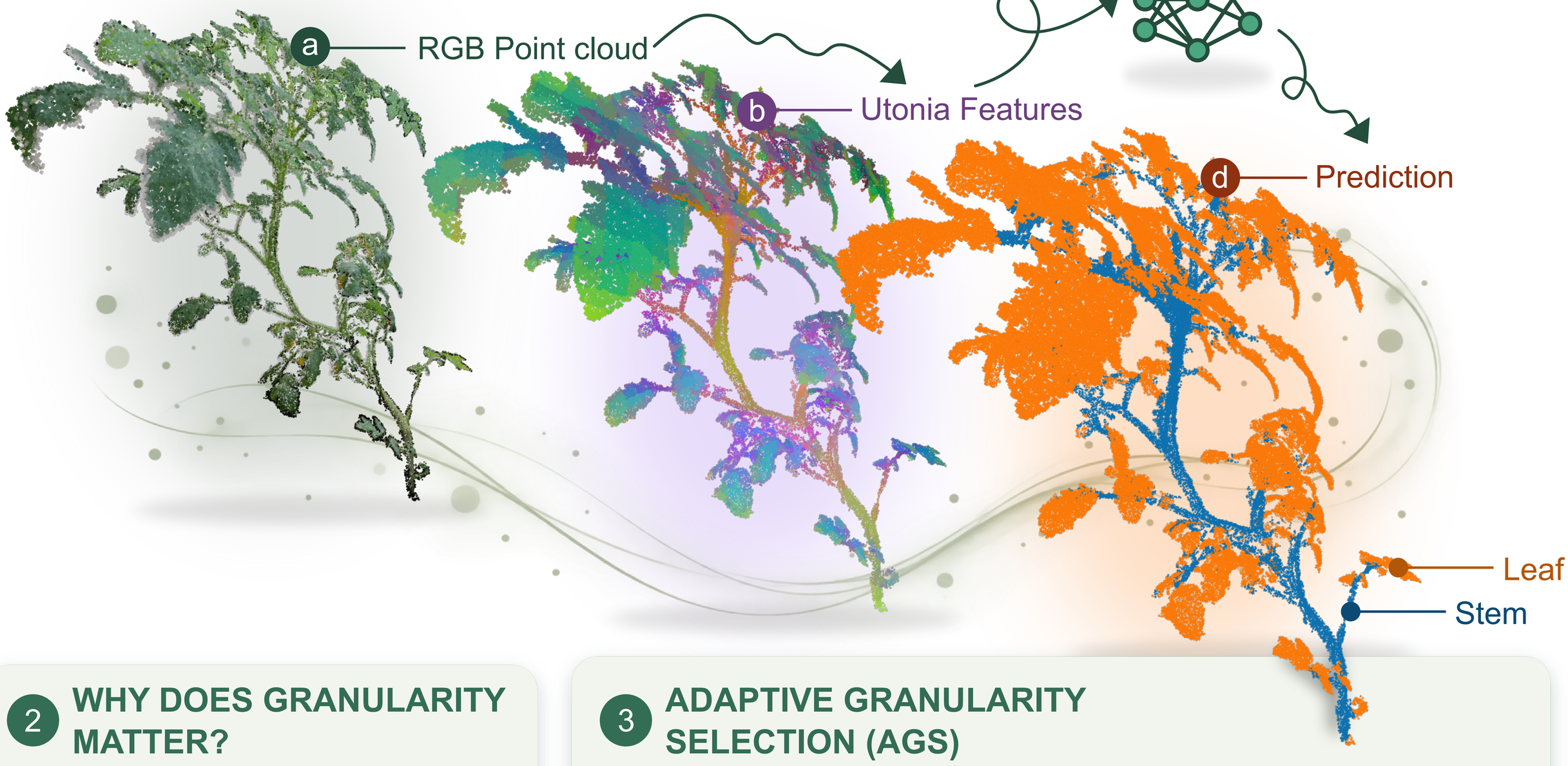
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ECCV 2026 | MALMÖ | SEPT 8–12

Webpage & Code

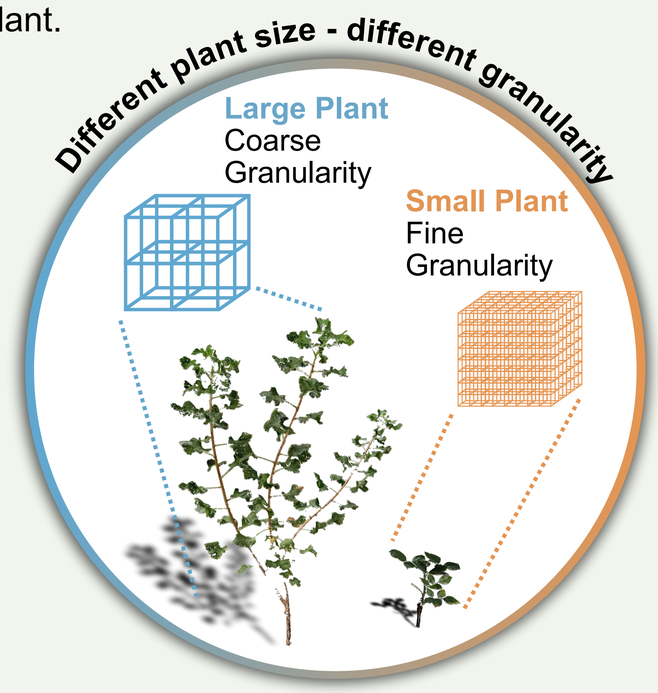
1 FROM 3D POINT CLOUDS TO PLANT ORGAN SEGMENTATION

We use a **foundation model** to extract point features, and train a lightweight **MLP head** to segment **leaf** and **stem** from 3D plant point clouds.



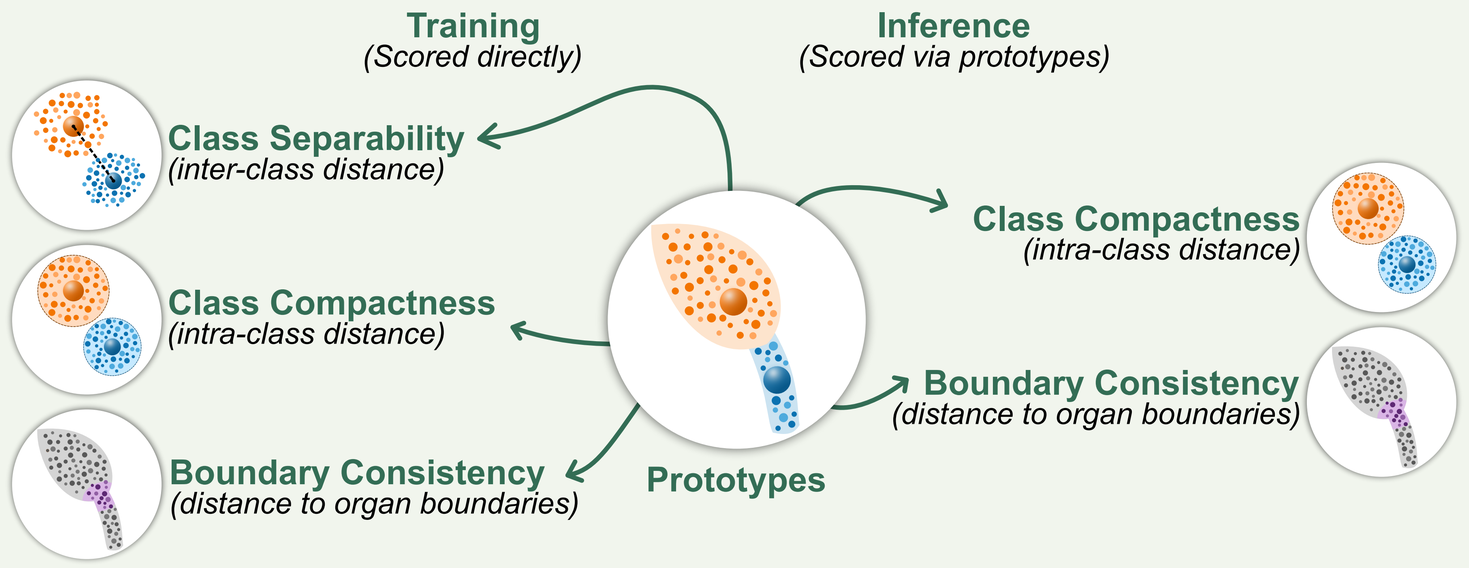
2 WHY DOES GRANULARITY MATTER?

Granularity: the resolution at which the plant is described. The best granularity differs from plant to plant.



3 ADAPTIVE GRANULARITY SELECTION (AGS)

AGS automatically selects **three granularities** per plant using **class prototypes**.
During **training**, selection prioritizes class discriminability and organ boundary consistency.
At **inference**, stored training prototypes guide selection, and predictions are combined through confidence-weighted aggregation.

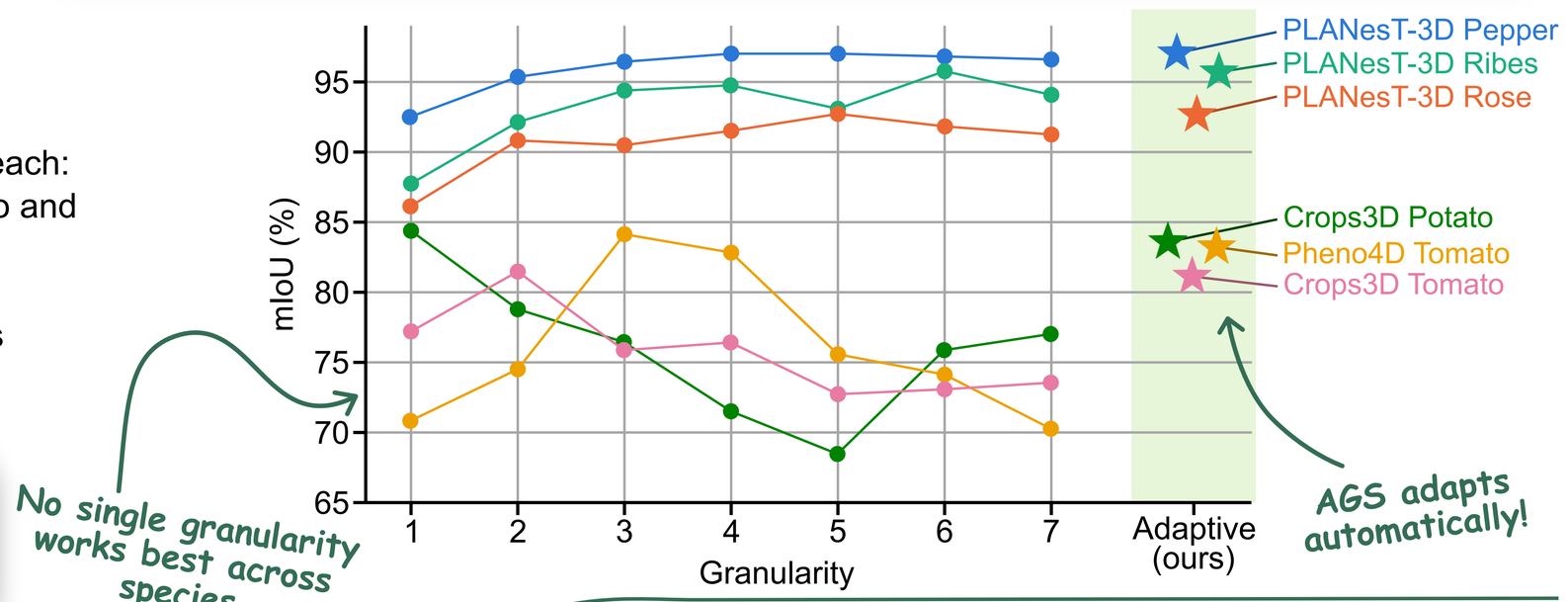


4 RESULTS ACROSS SPECIES

- **Few-shot generalization:** Trained on one plant each: pepper, rose, and ribes - tested on unseen tomato and potato.
- **Consistent performance:** 1st or 2nd best across species compared to fixed granularities

88.9%
Average mIoU
AGS-PlantSeg

+2.5
mIoU points
vs. best fixed granularity



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